

Human Review: Initial sign vectors in c_0

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct. It gives a simple counterexample to the question in Remark 5.9(iii) of Kurka's paper, under the statement as printed.

The example is

$$X = c_0,$$

and W is the union of the finite initial sign cubes

$$W_n = \{(\varepsilon_1, \dots, \varepsilon_n, 0, 0, \dots) : \varepsilon_i \in \{-1, 1\}\}.$$

The construction is very natural once the example has been found: the sets W_n are exactly the vertices of the finite cubes in the first n coordinates.

Convex Generation

The verification of

$$\overline{\text{co}}W = B_{c_0}$$

is correct. For each n , the convex hull of W_n is the cube

$$\{x \in c_0 : \text{supp } x \subseteq \{1, \dots, n\}, |x_i| \leq 1\}.$$

Therefore every finitely supported element of B_{c_0} belongs to $\text{co } W$. Since finitely supported elements are dense in B_{c_0} by coordinate truncation, the closed convex hull of W is the whole unit ball.

Weakly Convergent Sequences in W

The weak-sequential part is also correct. Suppose that

$$w_j \in W_{n_j}$$

is weakly convergent in c_0 . Weak convergence in c_0 implies coordinatewise convergence, because the coordinate functionals belong to $c_0^* = \ell_1$.

If the lengths n_j were unbounded, one could pass to a subsequence with $n_j \rightarrow \infty$. Then for every fixed coordinate k , the k -th coordinate of w_j is eventually always a sign, hence lies in $\{-1, 1\}$. Since this coordinate sequence converges, it must eventually be constant, and the weak limit would have

$$|x_k| = 1$$

for every k . This is impossible for $x \in c_0$.

Thus the lengths n_j are bounded. Consequently the sequence (w_j) takes values in a finite set. A finite-valued weakly convergent sequence in a Hausdorff topology is eventually constant, hence norm convergent.

Failure of the Schur Property

Finally, c_0 does not have the Schur property. The unit vector basis (e_n) is weakly null in c_0 , since for every $a = (a_n) \in \ell_1 = c_0^*$,

$$a(e_n) = a_n \rightarrow 0.$$

But $\|e_n\|_\infty = 1$ for all n , so (e_n) is not norm null.

Review Outcome

The proof appears correct. The construction is elementary, and the idea is clear once one notices the right set W : use initial sign vectors to generate the unit ball of c_0 , while forcing weakly convergent sequences in W to have bounded length and hence to be eventually constant.

This should be recorded as an apparently correct counterexample to the question as printed, subject only to the usual semantic check that no extra condition on W was intended.